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Descriptor: *Simon Fraser University Electric Vehicle Parking Dataset (SFU-EVP)*

STEPHEN MAKONIN ¹ (SENIOR MEMBER, IEEE), ISLA ZIYAT ² (MEMBER, IEEE),
RYLEN SAMPSON¹, SIYU AN¹, FRED POPOWICH ¹,
PATRICK PALMER ² (SENIOR MEMBER, IEEE), AND DAVID AGOSTI³

¹SFU's Big Data Hub, Simon Fraser University, Burnaby, BC V5A 1S6, Canada

²School of Sustainable Energy Engineering, Simon Fraser University, Surrey, BC V3T 0N1, Canada

³Parking & Sustainable Mobility Services, Simon Fraser University, Burnaby, BC V5A 1S6, Canada

CORRESPONDING AUTHOR: Stephen Makonin (e-mail: smakonin@sfu.ca).

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ABSTRACT Simon Fraser University (SFU) aims to make a significant contribution to the study of electric vehicle (EV) utilization and power grid management by providing a comprehensive dataset [Simon Fraser University electric vehicle parking dataset (SFU-EVP)] of EV charging sessions since 2019. This dataset will be continually updated in the future. This extensive dataset presents valuable information on EV charging patterns, providing critical input for power grid planning, policy development, rate design, and infrastructure placement. It also offers opportunities to improve load forecasting, ensure grid stability, and improve the integration of renewable energy. Furthermore, data can facilitate research toward optimizing various vehicle-to-grid (V2G) services, including harnessing EVs as distributed energy storage systems. All data are stored in the commonly used and easily accessible comma-separated value (CSV) file format. By making this dataset publicly available, SFU has created a vital dataset that can drive further innovation and efficiency in EV technology and grid management, fostering a more sustainable and environmentally friendly future.

IEEE SOCIETY/COUNCIL Power and Energy Society (PES)

DATA TYPE/LOCATION Time-Series; SFU Campuses, Metro Vancouver, Canada

DATA DOI/PID 10.21227/ya1w-m583

INDEX TERMS Charging, electric vehicle (EV), energy, power, sustainability, vehicle.

BACKGROUND

Public datasets on electric vehicle (EV) charging sessions can be extremely useful for various reasons related to power grid planning and stability, including load forecasting, infrastructure development, integration of renewable energy, rate design, vehicle-to-grid services, policy development, and phase balancing. In the province of British Columbia (BC), Canada, there are more than 150 000 EVs travel on BC roads [1]. Having comprehensive and reliable datasets on EV charging habits could significantly improve our ability to manage and plan for the growing prevalence of EVs.

This could lead to more efficient and sustainable power systems that can effectively handle increasing EV use to assist in both academic research and industry product/service development.

The continued adoption of EVs has prompted further inspection into the infrastructure and capacity of existing charging station networks to accommodate demand for these chargers, grid energy usage, and other limiting factors. This is most prevalent in large cities, high-traffic routes, or concentrated areas of business. Although data from charging stations are available, very few if any

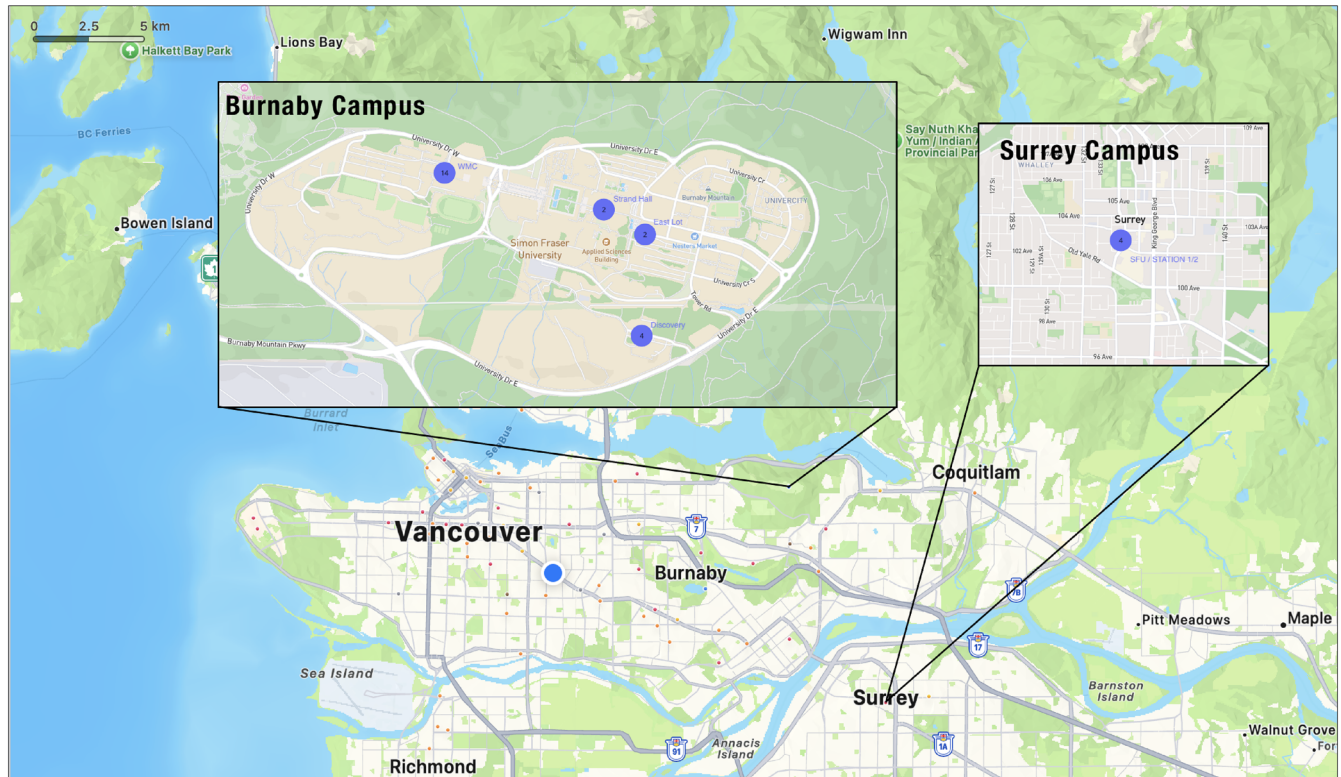


FIG. 1. Map of Metro Vancouver and each campus where EVs are installed. The zoom-in of the Burnaby and Surrey campuses shows the location and number of chargers installed at each location.

have compiled this data into an extensive dataset as we have done.

This is not the first dataset to study EV charging, but it offers unexplored avenues and perspectives, especially for Canada, where there is a lack of published data in this area. Some of these datasets are outdated and lack the volume representative of the current adoption [2]. Other datasets focus on residential EV charging, which has privacy implications, leading to private datasets or lack [3], [4]. ACN-data are more similar to our dataset in the aspect that EV chargers reside in a workplace environment [5]. Naturally, the locales of Caltech and JPL (ACN-data) are vastly different from SFU [Simon Fraser University electric vehicle parking dataset (SFU-EVP)] in terms of EV adoption, traffic, and so on.

COLLECTION METHODS AND DESIGN

The proposed dataset contains individual charging session data for EVs using EV charging station located on each campus, in the Metro Vancouver, BC, Canada area. Simon Fraser University (SFU) has a number of campuses in Metro Vancouver: the Burnaby Mountaintop campus (the largest and main campus), the city of Surrey, and Downtown Vancouver. Campuses with EV charging status installed are shown on the map in Fig. 1. For the Burnaby Mountaintop campus (the largest and main campus), the area includes university residences to the West, the main campus centrally, and a nonuniversity high-density residential area (called

UniverCity) of over 5000 people to the Northeast.¹ To the Southeast, there is a small area with university admin and Fraser International College (FIC). The Burnaby Mountain Parkway and Gagliardi Way also form a major route in the traffic system. Thus, the mountaintop is not as isolated as one might think when looking at the map.

The university boasts a combined student enrollment and campus employment of approximately 40 000 people as well as visitors who use available facilities such as gymnasiums and sports fields. This ensures it is a high enough traffic area to reasonably study EV charging trends, potential issues the increased demand will pose, and so on. The SFU campus covers roughly 2.0 km², with the charging stations dispersed in clusters across it. Regardless of the station, each charging session records common data (data collection depicted in Fig. 2).

Since its debut in 2019, the SFU charging station network has recorded 34 700 individual charging sessions that are included in the dataset at the time of publication. As these stations remain active for the foreseeable future, and more stations may be added, the dataset will periodically be updated with data. Figs. 3 and 4 both highlight the rate of new users that will have a profound impact on the number of charging sessions that can be expected in the future.

¹Read more about UniverCity at <https://university.ca/live-here/>

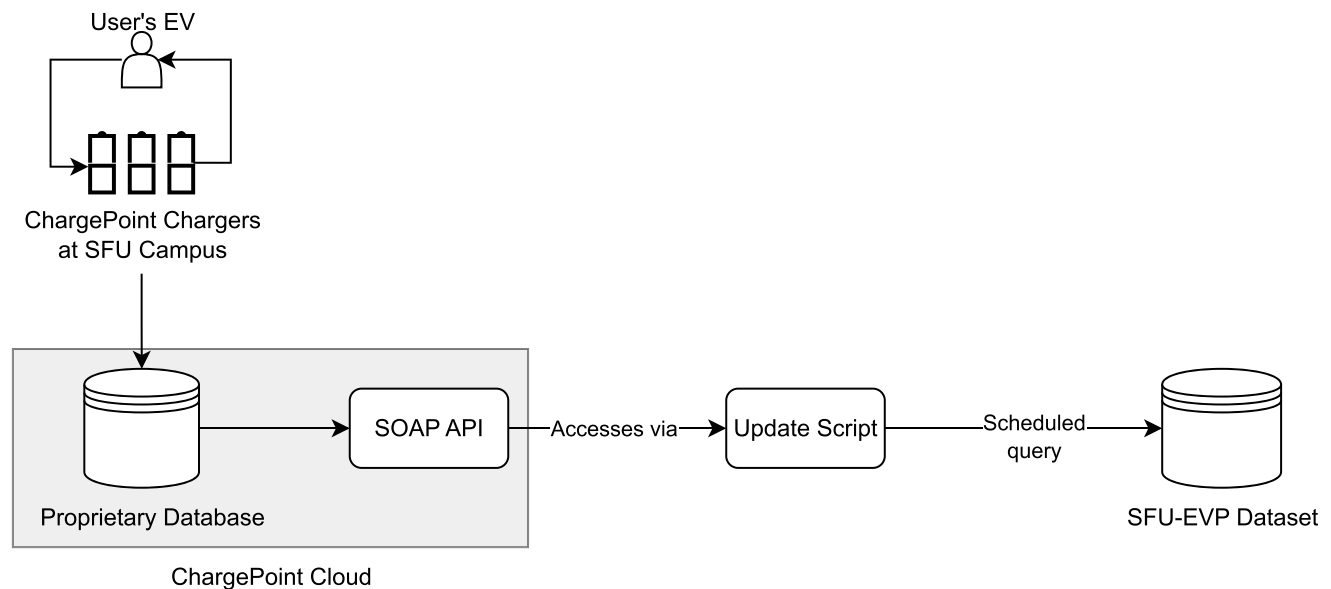


FIG. 2. Block diagram of the data collection system. ChargePoint chargers and their cloud solution are external to this research. Scripts to collect the data run on SFU virtual servers. While the Update Script runs, it converts the raw data, which includes removing and anonymizing data in a column that contains sensitive data. The dataset is hosted on IEEE DataPort.

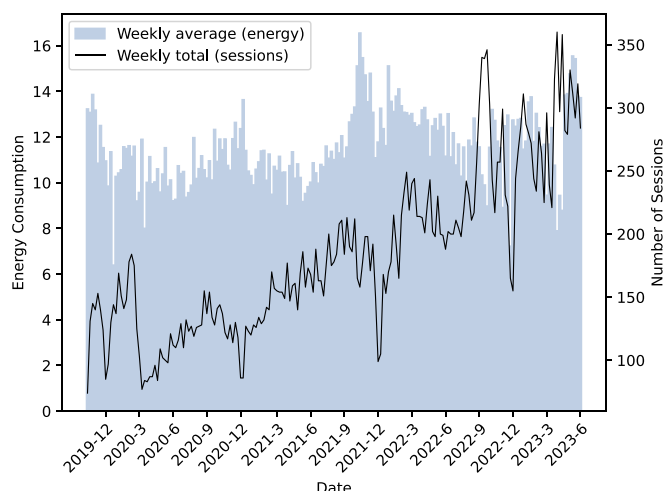


FIG. 3. Line and bar graph that displays the relation between an increasing number of charging sessions and the weekly average consumption of energy (kWh). The y-axis on the left side pertains to the bar graph, while the one on the right is for the line graph.

Furthermore, the dataset presented here includes details about each station within the EV charging network such as the number of ports available, voltage, power, and other hardware specs. Last, within the dataset are user data that contribute to understanding trends on that front.

Data Collection

Charging stations have been installed over a period of time starting in 2019. This enables monitoring of station usage as new stations are added and the evolution of user tendencies within a growing network. As seen in Fig. 3, the growing number of charging stations results in more

sessions, which in turn affects energy consumption. Rather than users recording their own data, the ChargePoint charging stations automatically document any sessions. This largely removes any potential human error or biases within the data. Additionally, any code written when collecting the ChargePoint data do not affect any facet of the data that would introduce errors. Note that this does not entirely prevent the possibility of erroneous data. Briefly, an error may occur when scenarios arise, such as one user unplugs another user's car prematurely. To address erroneous data, the alarms generated by the charging stations are included in the dataset. Details about possible alarms are covered in the *Records and Storage* section and their use cases in *Validation and Quality*. Furthermore, anomalous data are noted in a separate file from the charging session data and is further discussed in *Validation and Quality*. An example of anomalous data is one such user who left their car plugged in for more than 24 h.

All data are initially collected by the respective charging station, which then uploads the data to ChargePoint's cloud solution. This consists of a proprietary database and a simple object access protocol application programming interface (SOAP API [6]) that are used to interact with an organization's stored data. An external script can access the data to make specific queries and reorganize the data as necessary for an intended purpose. The initial data collection from the ChargePoint database through their API [6] was done with Python and in particular the Zeep library, which is a SOAP client [7]. For continued data collection, a daemon was written to run in the background on a regular basis that will append new charging sessions, stations, and user information to the dataset. The entire process is available in Fig. 2.

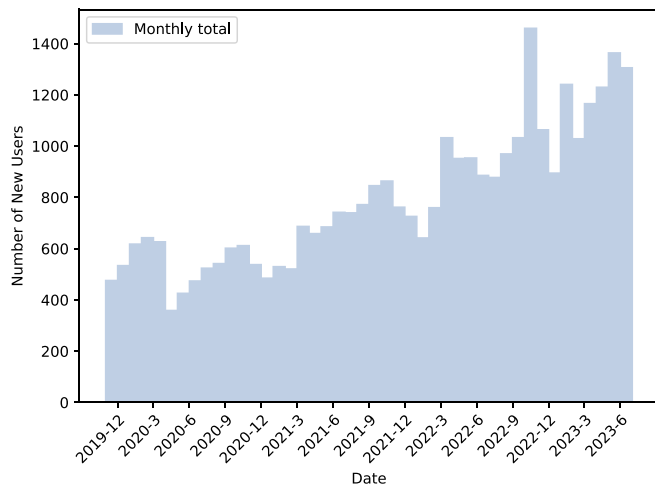


FIG. 4. Assessing the number of new users per month to the charging station network over the course of its existence. This is a proxy for understanding the adoption of public EV charging stations.

Pre-Processing and Removing User Data

The data required minimal data pre-processing as it was recorded automatically by the charging stations, and it is discrete data rather than continuous. Automated collection removes human error that arises during manual entry while discrete data offers last chances for machine failure when producing the final record. When a station is interrupted, ChargePoint will include an alarm indicating what affected the charging session. While the station's unique identifier is already linked to this information, it is pertinent to have the affected session as well. To facilitate linking and searching of data across various dimensions, the time of records is provided in both ISO and UTC format. The ISO date time format is provided at the local time of Burnaby, BC—Pacific Standard Time (PST). This is useful for drawing conclusions about how charging stations are used throughout the day as seen in Fig. 5. The ChargePoint database does not differentiate between ports at a charging station but does provide a way to do so. When pulling this data for the SFU-EVP dataset, each port is given its own row in the charging stations data as it will simplify energy and load analyzes.

The aforementioned pre-processing occurred during data collection. The main pre-processing done after data collection was the removal or obfuscation of identifying user information. Unique user and account identifiers were made anonymous to preserve the privacy of users within the network. This was accomplished by passing these identifiers through a hashing function. Any other identifiable information regarding users is removed such as the vehicle make and model.

Derived Variables and Their Maintenance

We augment our dataset with additional derived variables to further analyze and validate the existing data. Currently, sessions interrupted by hardware failures generate an alarm that is added to the corresponding data file. Originally, the alarms

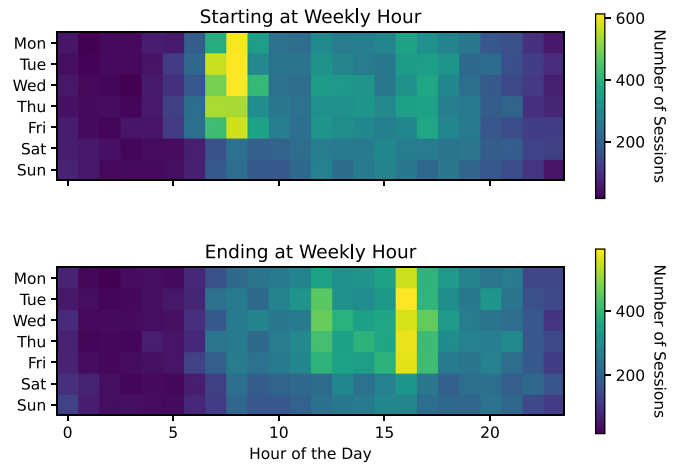


FIG. 5. Heatmaps showing the most frequent start (top) and end (bottom) times for each day of the week. On the x-axis, 0 pertains to 12:00 am PST and the final column is 23:00–23:59 pm PST.

data did not include the sessionID, which is counterintuitive as it would easily allow to search for charging sessions that were affected by some sort of hardware issue. We incorporate this addition by finding the charging session that includes the time of the alarm in its start and end times. Conversely, anomalous data do not generate an alarm but should be noted. We provide an entirely separate data file for anomalies that lends itself to analyzing user behavior and what could be considered poor EV charging etiquette. This data file contains the following: the ID of the anomalous session; a description of the anomaly, the formulation of anomalies is discussed in *Validation and Quality*; the anomalous value; and the respective unit of the value.

Maintenance of these additional variables is straightforward. Including the sessionID in the alarms data file only requires running a simple Python function for the process described above. Similarly, to check for anomalies, all that is needed is to pass the data to a function that looks for any data that fails certain conditions. Furthermore, only newly collected data needs to be checked, which means the runtime moving forward will be minimal.

VALIDATION AND QUALITY

The scripts stored in the GitHub repository offer the capability to reproduce all the statistics and figures presented in this article. By utilizing the scripts, researchers or readers can execute the necessary computations and generate the statistical results and visualizations mentioned in the article. This ensures transparency and allows others to independently verify and validate the findings and conclusions presented in the research.

Continual Updates

Our programs designed for downloading and uploading operate as daemon processes to continually update the dataset at regular intervals of 24 h. Log files are provided to allow users to track the programs' execution history and any significant

TABLE I. Description of Charging Session (Sessions.csv)

Feature	Type	Description
session_id	String	Unique identifier assigned to each charging session.
user_id	String	Unique identifier assigned to each user of the charging session.
credential_id	String	Unique identifier associated with the credentials/authorization used by the user during the charging session.
station_id	String	Unique identifier assigned to the charging station where the session took place.
port_no	Integer	Port number/identifier associated with the specific charging port at the charging station.
start_ts	Integer	Starting timestamp of the charging session in UTC time and seconds format.
end_ts	Integer	Ending timestamp of the charging session in UTC time and seconds format.
start_dt	DateTime	Local datetime (YY-MM-DD HH:MM:SS) of the session starting.
end_dt	DateTime	Local datetime (YY-MM-DD HH:MM:SS) of the session ending.
energy	Float	Amount of energy (in kWh) consumed or delivered during the charging session.
total_charging_duration	String	Total duration for which the current was flowing during a session.
total_session_duration	String	Total duration of the session; until users disconnect their vehicles from the charging ports.
address	String	The physical address/location of the charging station where the session took place.

updates. The *Sessions.log* file contains information about the time range of a query and the data size before and after merging. Upon program execution, it first checks for the existence of the designated *Sessions.csv* file. If the file is not found, a default query is performed with a time range from 1 January 1970 at 00:00:00 to the current UTC timestamp, and the resulting data are saved to the specified path. If the historical data exist, a query is made with a time range from the end time of the last recorded entry to the current UTC timestamp, and the retrieved data are merged with the existing dataset. The log of alarms shares similarities with sessions. Additionally, the *Stations.log* file provides information about the number of records and the file path where the station data are saved. It is important to note that the program executes a query every 24 h without verifying the file's existence. Consequently, the data file is overwritten with each query completion, ensuring that only the most up-to-date information is retained.

Privacy

To address privacy concerns, a hash function is applied to anonymize the *user_id* and *station_id* attributes in the sessions data, as well as the *station_id* and *org_id* attributes in the stations and alarms data. This hash function employs encryption techniques to transform the original values, safeguarding the identities of users and stations. Moreover, it is essential to highlight that the uploaded files are replaced with each dataset update, guaranteeing that only the latest and accurate data are accessible for analysis.

Anomalies

Extreme cases where certain users have charging sessions lasting up to 19 529 min exist in the raw dataset. A pre-processing step was hence undertaken to remove outliers based on the duration of charging sessions prior to exploring the dataset. This was achieved by calculating the duration of each charging session in minutes, obtained by subtracting the start timestamp from the end timestamp. The resulting

durations were then used to determine the first and third quartiles of the distribution. The interquartile range (IQR) was calculated as the difference between the third and first quartiles. Subsequently, lower and upper bounds for outliers were defined using 1.5 times the IQR. Rows in the original dataset where the duration fell outside the range specified by the lower and upper bounds were filtered out, resulting in a new dataset devoid of outliers. This pre-processing step allowed for a more focused and accurate exploration of the dataset by eliminating potential anomalous or extreme values.

Level 2 charging is typically arranged with EV parking, unlike fast charging. In addition, level 2 EV charging is usually considered as appropriate for charging and *topping off* [8] the charging to get to 100% charge at a slower rate under the control of the vehicle's battery management system. Therefore, a low rate of charging and even pulses of charging cannot be considered as anomalous [9].

RECORDS AND STORAGE

The dataset provided consists of four comprehensive data files, containing detailed information regarding the charging sessions, stations, alarms raised on stations, and anomalies. These files are stored in the comma-separated values (CSVs) format, enabling structured and organized representation of the data.

In the *Sessions.csv* file, each row represents a complete charging session, encompassing the entire process of charging an EV. The session starts when the EV is connected to the charging port and concludes upon reaching the desired or full charge. This file enables an understanding of the duration, energy consumption, and other session-specific details. Table I describes each column.

The *Stations.csv* file contains comprehensive information about the charging stations. In cases where a station consists of two ports, the data are structured to represent each port as a separate entry. As a result, for stations with dual ports, two

TABLE II. Description of Station (Stations.csv)

Feature	Type	Description
station_id	String	Unique identifier assigned to each charging station.
org_id	String	Unique identifier representing the organization or entity associated with the charging station.
station_group	Integer	Custom station group identifier to help organize and classify charging stations into distinct groups.
model	String	Charging station manufacturer model number.
activation_dt	DateTime	The local date and time when the charging station was activated or made operational.
timezone_offset	DateTime	The time offset from UTC for the charging station's specific time zone.
address	String	The physical address or location of the charging station.
manufacturer	String	The manufacturer or brand of the charging station equipment.
station_name	String	The name or label assigned to the charging station.
description	String	A brief description or additional information about the charging station.
port_no	Integer	The port number or identifier associated with a specific charging port at the charging station.
reservable	Boolean	Indicates if the charging station can be reserved or booked in advance: "1" for YES and "0" for NO.
status	String	Current operational status of the charging port: available, offline, or undergoing maintenance.
level	Enum	Refers to the station level type, such as Level 1, Level 2, Level 3, or Level 4 (DC Fast).
time_stamp	Integer	A timestamp associated with the charging port.
mode	Enum	The station mode type: Mode 1/Mode 3/Mode 4 (DC Fast). For EU Stations only, else use "Level".
connector	String	Describes the type of connector or plug used for charging EVs at the station.
voltage	Float	The voltage (in V) level at which the charging station operates.
current	Float	The electric current (in A) provided by the charging station during the charging process.
power	Float	The rated power (in kW) output or capacity of the charging station, typically measured in kilowatts.
estimated_cost	Float	An estimation of the cost.
location_lat	Float	The latitude coordinate of the charging station's location.
location_long	Float	The longitude coordinate of the charging station's location.

distinct records are available in the file, distinguished by their respective *port_no* values while sharing the same *station_id* value. This approach allows for a more comprehensive representation of the station's attributes and facilitates individual management and analysis of charging ports within a station. Table II describes each column.

The *Alarms.csv* file is dedicated to capturing and documenting the alarms raised at charging stations. Each record within this file corresponds to a specific alarm occurrence observed during a charging session. Notably, some entries in the *port_no* feature, which denotes the port number at the station, contain missing values. These omissions indicate a potential scenario wherein the alarm pertains to the entire station rather than a specific port. Furthermore, the *session_id* feature, not directly obtained from the API, is derived by cross referencing the records in the sessions data. This process entails matching the *station_id* and *port_no* values, followed by an examination of the *alarm_ts*, a UTC timestamp indicating the alarm triggering time, within the time interval defined by the *start_ts* and *end_ts* columns in the sessions table. Instances where the *session_id* feature exhibits missing values signify unmatched records between the alarms and the corresponding sessions. Table III describes each column.

Stored in the *Anomalies.csv* file are instances of charging sessions that exhibit irregular charging patterns. Each observation in the file corresponds to a specific charging session, uniquely identified by its *session_id* followed by

supplementary details, including categorization information, magnitude, and measurement units, for each recorded session. Table IV describes each column.

The file names in the dataset are designed to provide descriptive information about the data they contain. For example, the *Sessions.csv* file specifically contains charging session data. By referring to the file names, users can easily identify and differentiate the specific data they require.

INSIGHTS AND NOTES

The dataset utilized in this study captures the charging histories spanning from 6 November 2019 to 3 July 2023. It is important to note that the dataset is subject to regular updates, leading to an expanding coverage over time. Within this dataset, there exist a total of 34 944 observations of charging sessions, all of which are contributed by 2155 distinct users, providing a diverse range of charging session data for comprehensive analysis.

The stations data within the dataset comprises 13 charging stations, each equipped with two charging ports. To accommodate this configuration, the data are structured in a manner where each charging port is presented as a separate row, resulting in a total of 26 records. This approach enables a detailed examination of the individual charging ports within each station, facilitating a comprehensive understanding of the station infrastructure.

The historical alarms include a total of nine relevant features and data from 6 November 2019 to 13 July 2023.

TABLE III. Description of Alarm (Alarms.csv)

Feature	Type	Description
station_id	String	A unique identifier assigned to each charging station.
station_name	String	The name or label assigned to the charging station.
model	String	Station manufacturer model number.
org_id	String	An identifier representing the organization or entity associated with the charging station.
port_no	Integer	The port number or identifier associated with a specific charging port at the charging station.
alarm_type	String	The type of the alarm.
alarm_ts	Integer	The UTC timestamp of when the alarm occurred.
alarm_dt	DateTime	Local datetime (YY-MM-DD HH:MM:SS) of when the alarm occurred.
session_id	String	A unique identifier assigned of the charging session.

TABLE IV. Description of Anomaly (Anomalies.csv)

Feature	Type	Description
session_id	String	This is a unique identifier allowing for referencing of data points pertaining to specific charging events.
anomaly_description	String	A label that describes the detected anomaly to help identify/categorize different irregular charging patterns.
value	String	A measurement of the magnitude/severity of the deviation from expected charging behavior.
unit	String	It indicates the specific measurement unit corresponding to the "value" column.

TABLE V. Descriptive Statistics

Feature	Min	1st Quantile	Median	3rd Quantile	Max	Mean	SD
Energy consumption (kWh)	0.00	2.93	7.24	14.51	57.36	10.28	10.14
Duration (minutes)	0.15	42.27	124.72	222.11	532.57	147.71	125.96

The size will also grow as the dataset continues to accumulate new data.

Exploratory Analysis

The analysis performed in this study utilizes a dataset spanning from 6 November 2019 at 15:49:00 UTC to 3 July 2023 at 15:37:36 UTC. A pre-processing stage was undertaken to eliminate outliers from the dataset before exploring.

During our exploration of the dataset, we conducted analyses on charging sessions, focusing on energy consumption and duration. Additionally, we examined the utilization of the charging station at different times of the day. Table V presents descriptive statistics that shed light on the observed patterns. Regarding energy consumption, the data reveal a range of values from 0.00 kWh to a maximum of 57.36 kWh. The majority of sessions tend to have energy consumption below 14.51 kWh, as indicated by the third quartile. On average, sessions have an energy consumption of 10.28 kWh. The standard deviation of 10.14 is close to the mean and suggests a moderate level of consistency in energy consumption across sessions. However, it is worth noting that some sessions exhibit no energy consumption. Turning to session duration, the statistics illustrate a diverse range of values. The minimum recorded duration is 9 s, indicating extremely short sessions. The median duration stands at 124.72 min, with half of the sessions falling below this value. The mean duration is 147.71 min, providing an estimate of the typical session length. In contrast to energy consumption,

the duration statistics reveal a wider range of values, with a standard deviation of 125.96. This suggests substantial variation in session lengths.

Fig. 6(a) visually represents the distribution of energy consumption during charging sessions, indicating that a majority of users have energy consumption levels below 10 kWh. This suggests that users tend to charge their vehicles for shorter periods or have lower energy requirements. The histogram includes two curves representing the density and exponential distribution, implying a potential trend of energy consumption following an exponential distribution pattern. Additionally, the histogram Fig. 6(b) shows that the charging sessions typically lasted between 1 and 3 h. Similarly, the two line charts, representing the density and expected distribution, visually resemble an exponential distribution trend.

Fig. 7(a), from a different perspective, presents the distribution of station usage using two distinct curves. The start time of charging sessions is represented by the red curve, while the blue curve illustrates the end time of these sessions. Examining the start and end curves reveals a discernible trend: a majority of sessions occur during the daytime, particularly falling between 6:00 am and 6:00 pm. The curves exhibit two prominent spikes, indicating users' proclivity to initiate charging around 8:00 am and wrap up their charging activities by approximately 4:00 pm. This fact strongly suggests that a significant cohort of users prefer utilizing the charging stations within close proximity to their

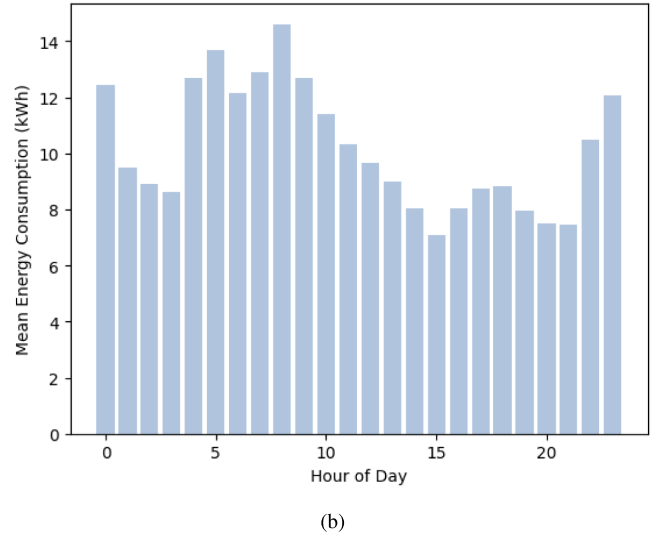
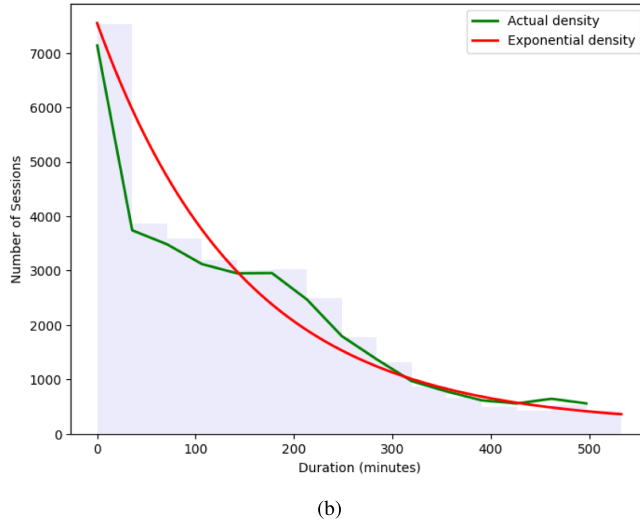
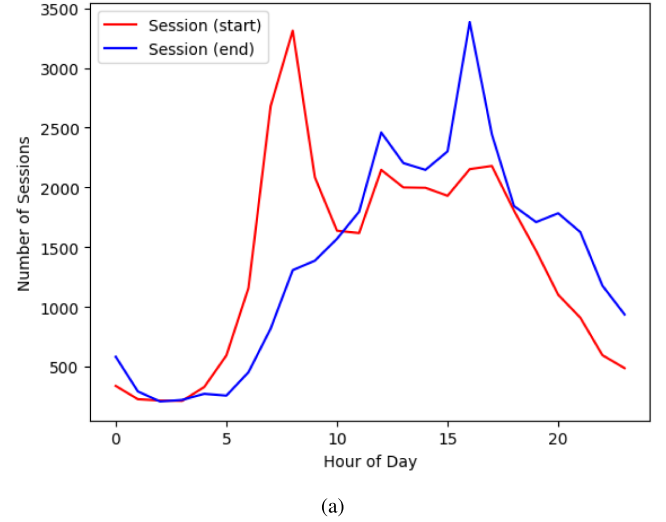
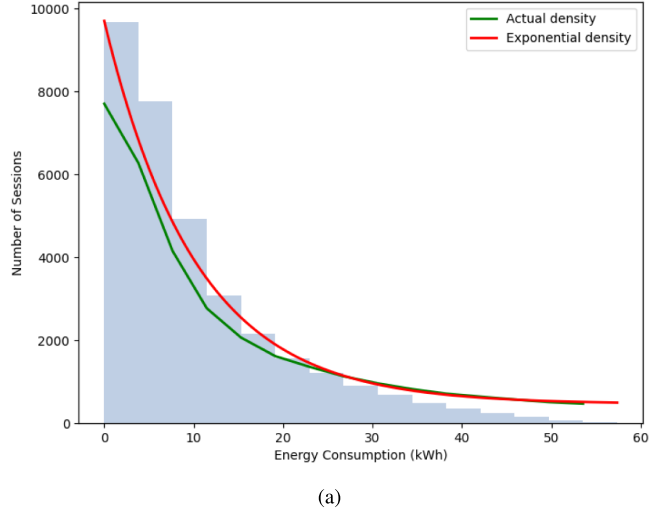


FIG. 6. (a) Energy consumption and (b) charging duration versus the number of sessions.

FIG. 7. Station and energy usage where *hour of day* is local time (Pacific timezone). (a) Station usage by hour. (b) Mean energy usage by hour.

workplaces. This same observable pattern is also displayed in Fig. 7(b), which graphically represents the average energy consumption. The methodology involves aggregating the energy expended within each discrete time interval and subsequently calculating the mean. Notably, the pinnacle of energy consumption occurs at 8:00 am, followed by a dip until 3:00 pm, with the corresponding energy utilization being quantified at 14.61 and 7.07 kWh, respectively. This underscores a conspicuous demand for charging services during these specific temporal windows. Interestingly, there is an additional surge in consumption evident from 10:00 pm to 12:00 am, closely resembling the levels of utilization witnessed during the daylight hours. This might be due to the residents living on campus seeing online that the EV parking stall is free, and they would like to charge before heading down the mountain the next day.

Fig. 8 shows the daily total charging power of all the chargers for each day in 2022. It is clear, based on the

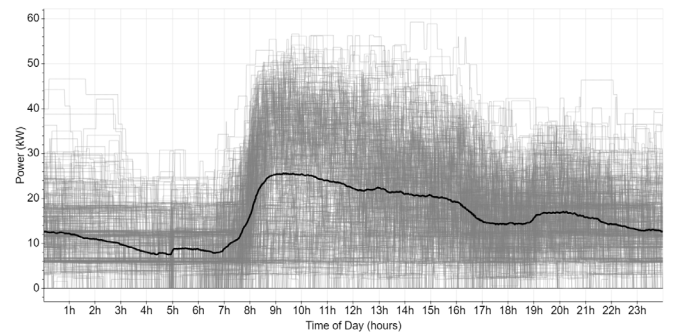


FIG. 8. Daily total charging power for each day in 2022, showing the average charging power in the thick black line.

average charging profile, that peak charging occurs around 9:00 am as people arrive to work on campus, and then power gradually decreases throughout the day as the EVs complete their charge. There is a clear dip at 5:00 pm when

people depart from work and a small increase at 7:00 pm when residents arrive home. The SFU campus is a unique case study since EV users can work and/or live on campus; therefore, there is always a base charging of around 10 kW through the evenings and overnight from residents who choose to charge during that period. The lowest charging period is between 4:00 and 7:00 am, when the overnight users complete their charging.

Limitations and Future Work

Given the low sampling rate of this dataset, it would be hard to match the profiles of different EV makes/models/year algorithmically, similar to the *nonintrusive load monitoring* (NILM) research we have done in the past [10]. We would hope to eventually collect data at either 1 Hz or per-minute sampling rates. This would allow us to characterize the detailed charging profile and use the high-frequency data collected in the *EV charging profiles and waveforms* (EV-CPWs) dataset [8] to identify the make/model of the vehicle once down-sampled to 1 Hz.

SOURCE CODE AND SCRIPTS

All scripts utilized for generating the dataset and figures presented in this article have been made publicly available through a GitHub repository. To access the scripts, please follow the link provided: <https://github.com/sfu-bigdata/EVDataset>. Note that these scripts can be used for any ChargePoint station installed at the cloud portal.

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S. Makonin reviewed the curation and wrote parts of the manuscript. I. Ziyat, R. Sampson, and S. An curated and analyzed the data and wrote parts of the manuscript. F. Popowich, P. Palmer, and D. Agosti provided feedback and edits. All authors reviewed the manuscript.

The authors have declared no conflicts of interest.

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